

High-level language programming project--specification

1. Objectives

The goal of this course is to practice the students' capability of problem analysis and programming skill.

- 1) To master the thinking of **object-oriented programming**, and use idea of object-oriented programming to solve practical problems;
- 2) To carry out the programming design and coding **independently**, and use various ways for debugging and correcting the error efficiently;
- 3) To write scientific **report**, and hence have a better understanding of the knowledge and to develop students' ability to express and speculative.

2. Requirements

- 1) **Every 6-8 students can be grouped** together to do the project.
- 2) Each group of students selects **one interesting topic which is not limited to the options provided** in this specification. Finally, each group should submit the proposal (appendix 1) with word number not less than **1000**, project report (appendix 2) with word number not less than **3000** and source code;
- 3) Techniques:
 - a) The key features of the object-oriented programming language should be utilized, including class, *objects*, *encapsulation*, *inheritance*, *polymorphism*, etc. ;
 - b) The number of class should not be less than 5;
 - c) The number of levels for the inheritance hierarchy should not be less than 2;
 - d) The UI is needed for the system;
 - e) Random file processing (writing, reading, and updating) is also required in the system;
 - f) The lines of source code should not be less than **2000**.
- 4) The system should be designed and implemented reasonably, efficiently and friendly.
- 5) Source code should be clearly commented: comments for class (a detailed description of the properties, behaviors, class inheritance, and derived relationships); comment for functions (detailed description of objectives, function parameter, principle steps /algorithms).

3. Reference topics

● Dibuix

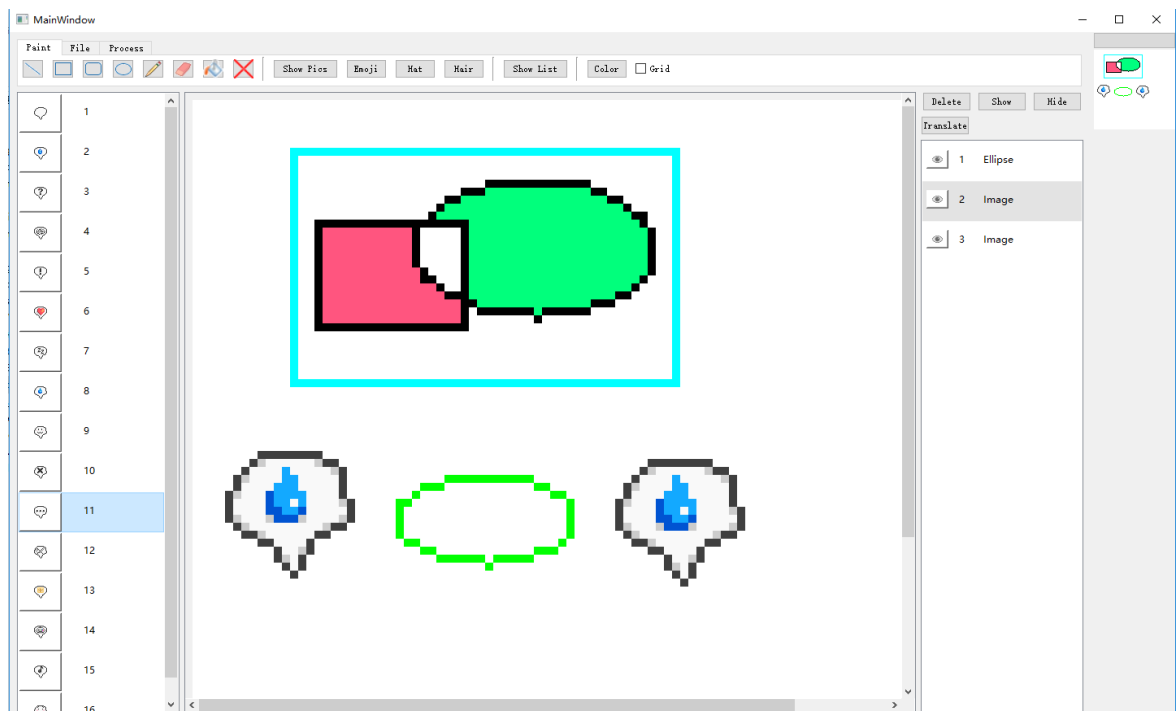
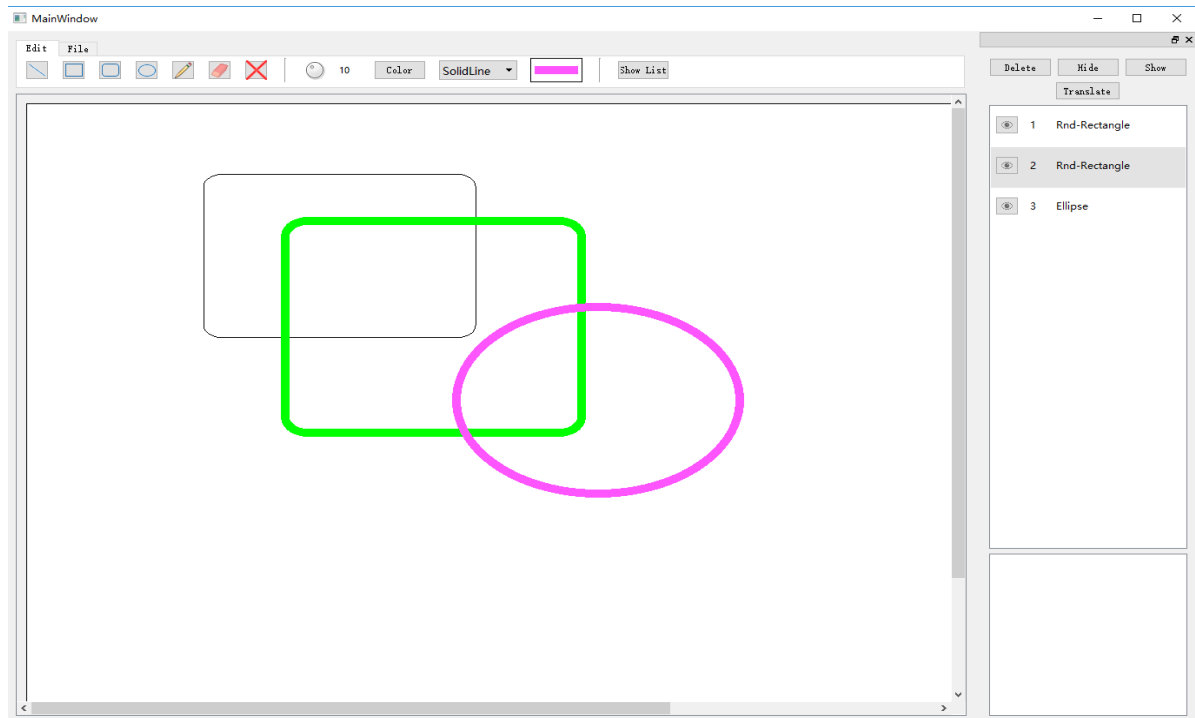
Description: Dibuix is a Catalan word which means drawing, so literally, it is a painting application, but can do more than painting. First, as a drawing software, it will still include the basic drawing function components to meet the basic drawing needs of users. Moreover, in order to make the software design novel and innovative, and conform to the characteristics of the AI era, we will provide functions like face recognition (may even include body

recognition) so the software can automatically draw a painting. After uploading a picture, the software can recognize the picture intelligently and draw the people inside it with in pixel style.

Request:

1) Provide a friendly UI for the system;

Hints: MFC, QT, C sharp can be used;



2) Wizard dialog box for files operations

3) Normal painting mode:

- Basic shapes drawing
- Clear canvas
- Pen's demo canvas
- Adjust pen's properties
- Visualize shape list
- Hide, show and delete shape
- Translate shape
- Visualize adjust list
- Outline shape's range
- Save and open picture

4) Pixel art mode

- Basic shapes drawing
- Clear canvas
- Adjust pen's properties
- Visualize shape list
- Hide, show and delete shape
- Translate shape
- Change previous shape's color
- Fill shape
- Existing images for directly drag and draw
- Grid mode
- Gray mode
- Fill shape
- Save and open picture
- Save and open project file
- Adjust brightness
- Adjust saturation
- Actual-size demo canvas

● Optional subject

- 1) Bank Account Management System
- 2) Note-taking and Schedule Tool
- 3) Monopoly
- 4) Supermarket Management System
- 5) Household registration information management system
- 6) Personal Financial Management System

Or other topics which have similar difficulty.

Note: The proposal for the selected topics should be submitted before the determination of the topics for the project.

4. Evaluation Criteria

Full score is 100 points. The details are as the following.

1) Source Code-----60%

- Basic requirements-----55%

- ✓ correctness-----30%

Demonstrate the system using complete and sufficient test cases; All of the output are correct.

- ✓ Rationality of design-----10%

The design of classes and inheritance hierarchy should be able to solve the key issues of the system.

- ✓ User Interface -----15%

Use Friendly and have nice visual effect.

- Additional features for refinement -----5%

2) Demonstration-----20%

- ✓ Expression clarity-----10%

- ✓ Performance on answering questions -----10%

3) Final report -----20%

Requirements: Refer to the appendix 2.

Requirements: clearly express the design ideas, the technical highlights, the final result and experiences. Demonstrates the program and able to answer the questions clearly and reasonably.

Note: If more than two or above students submit the similar or even same source code and report, after careful verification, the students who copy the files from others will be given zero score. **Plagiarism is not permitted!**

5. Submission File

1) Proposal

2) Final report (electronic version)

3) Source code and executable programs

Appendix 1: The template for proposal

一、 Requirement Analysis for System

1.1 The Background and Motivation of System

1.2 System Objectives

二、 Program Analysis

2.1 Key Issues for System

2.2 Duty Assignments

The description of duty for every student in the group is listed here.

三、 Reference

Appendix 2: the template of the report



华南理工大学

South China University of Technology

《High-level Language Programming Project》 Report

Project Name: _____

School : _____

Major : _____

Student Name : _____

Teacher : _____

Submission Date: _____

1. Requirement Analysis for System

1.1 The Background and Motivation of System

1.2 System Objectives

2. Program Analysis

2.1 Key Issues for System

2.2 Duty Assignments

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3. Technical Routine

3.1 Runtime Environment

The platforms of software and hardware for the system are described here.

3.2 General Design

3.3 Detailed Design

4. Programming Progress

Phases of Mission	Period	Planned Completion	Actual Completion
Determining the topic and submit the proposal	2021.04.26-2021.05.03	Survey the interesting topics, and writing the proposal.	

Note: The progress for each week should be recorded. The project will be started from April 26, 2016 and ended at June 7, 2016.

5. Testing Report

5.1 Function testing

5.2 System testing

6. Personal Summary

Write down your acquisition, suggestions, etc..

7. Reference